6. Project Testing

6.1. Testing Methodology

1.Dataset Testing – The dataset is checked for missing values, errors, and data consistency.

2.Model Testing – The trained machine learning model is evaluated using test data.

3.Performance Testing – Metrics such as Accuracy, Precision, Recall, and F1-score are used to measure model performance.

4.Input Testing – Different customer details are provided to verify correct prediction outputs.

5.Validation Testing – The model results are compared with actual outcomes to confirm reliability.

6.2. Automated Test Configuration (`test_app.py`)

The automated testing setup validates the performance and functionality of the credit card approval prediction system.

1. Configure the testing environment with required Python libraries and dependencies.
2. Load the trained machine learning model and test dataset.
3. Execute automated test cases with different input values.
4. Compare predicted results with expected outcomes.
5. Generate test reports using evaluation metrics such as accuracy, precision, recall, and F1-score.

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6.3. Test Results Table

Test Case ID	Test Component	Input Payload	Expected Output	Actual Output	Status
TC-01	Home Route Loading	GET `/`	Status 200, contains "	:rediStatus 200, text erifie Shield AI" c	PASS
TC-02	Form Page Loading	GET `/predict`	Status 200, contains "	·pplicant Profile"Status 00, text verified	PASS
TC-03	Approval Prediction	Age=35, Income=120k,	Debt=0, FICO=820Page renders "Approv	·d"Page renders Approv	·d"PASS

TC-04	Rejection Prediction	Age=22, Income=15k,	Debt=20k, FICO=350Page renders "Decline	d"Page renders 'Decline	d"PASS
TC-05	Boundary Validation	FICO=900 (Invalid)	Renders: "Credit Scor	must be 300-50"Renders validation var	ningPASS

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